

What is generative media?

Generative AI

AI that generates new content

e.g.

text generation

code generation

synthetic data

Generative Media

Generative AI applied to media

e.g.



Image



Video



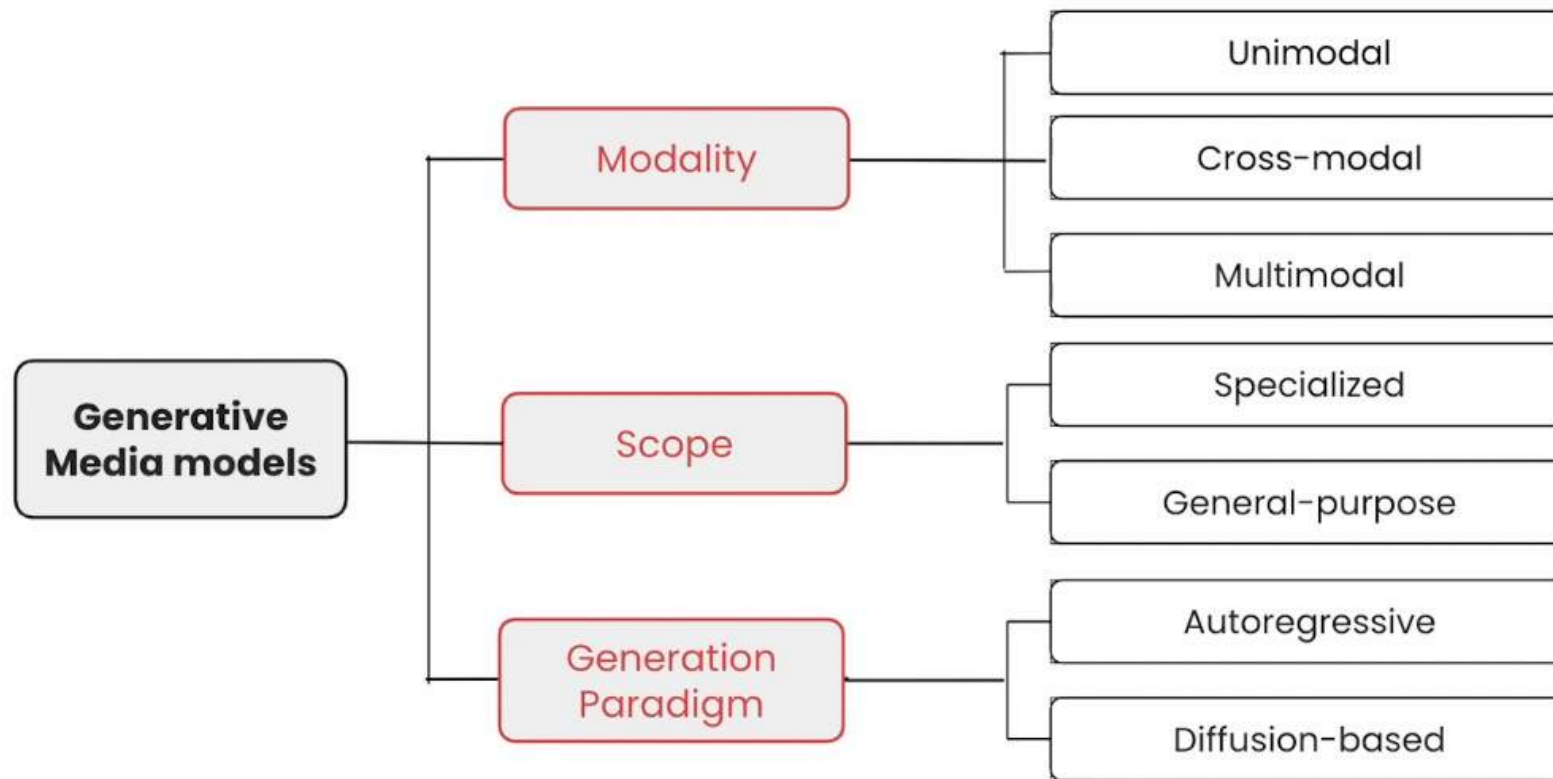
Audio



Music

Generative media models

Three dimensions for understanding how generative media models work



Generative media: modalities

Generative media models can operate on a single modality or across multiple modalities

UNIMODAL

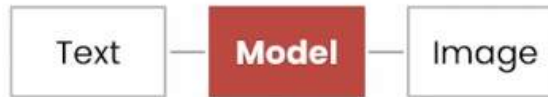
Same modality in and out



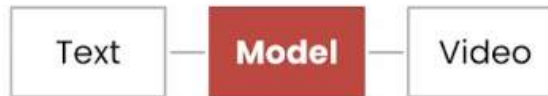
e.g. image upsampling, super-resolution

CROSS-MODAL

One modality in, different modality out



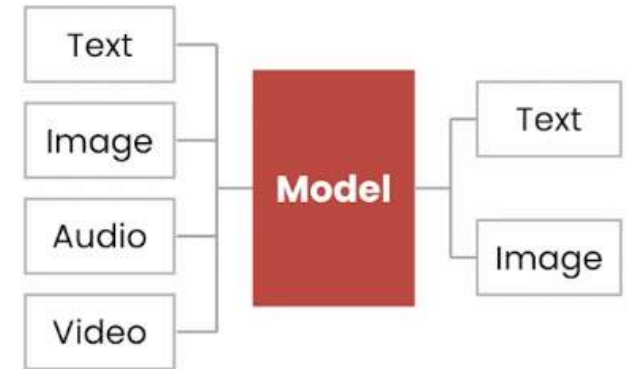
e.g. image generation



e.g. video generation

MULTIMODAL

Multiple modalities in and/or out



e.g. text, image, audio, video in → text, image out

Generative media: scope

How generative media models differ in scope

Specialized models

Focused on one media type

e.g.



Lyria

Music generation



Veo

Video generation with synchronized audio

General-purpose models

Work across multiple media types

e.g.



Gemini

Understands text, image, video and audio.
Generates text, image and audio.

Generative media: generation paradigms

Two paradigms that modern models increasingly combine

Autoregressive

Generates media sequentially, one element at a time, image patches, audio frames, or video tokens. Each element is conditioned on everything generated before it.

Excels at: Fine-grained control, instruction following, and compositional accuracy.

Diffusion-based

Starts from random noise across the entire output, image, video frame, or audio signal, and gradually refines it into a coherent result through iterative denoising steps.

Excels at: High-fidelity outputs, rich detail, and visual or sonic coherence.

Generative media: autoregressive generation

Illustrative example of a generic autoregressive architecture for generative media

1 Encode

Each input modality passes through a tokenizer, unified or modality-specific depending on the model.

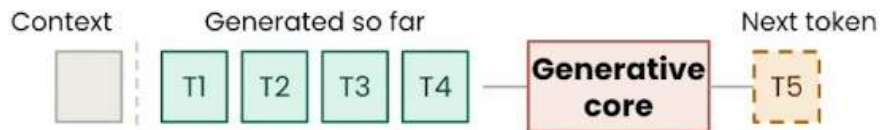
2 Generate

The generative core predicts the next output token conditioned on all prior tokens. This loop repeats until the full output is built.

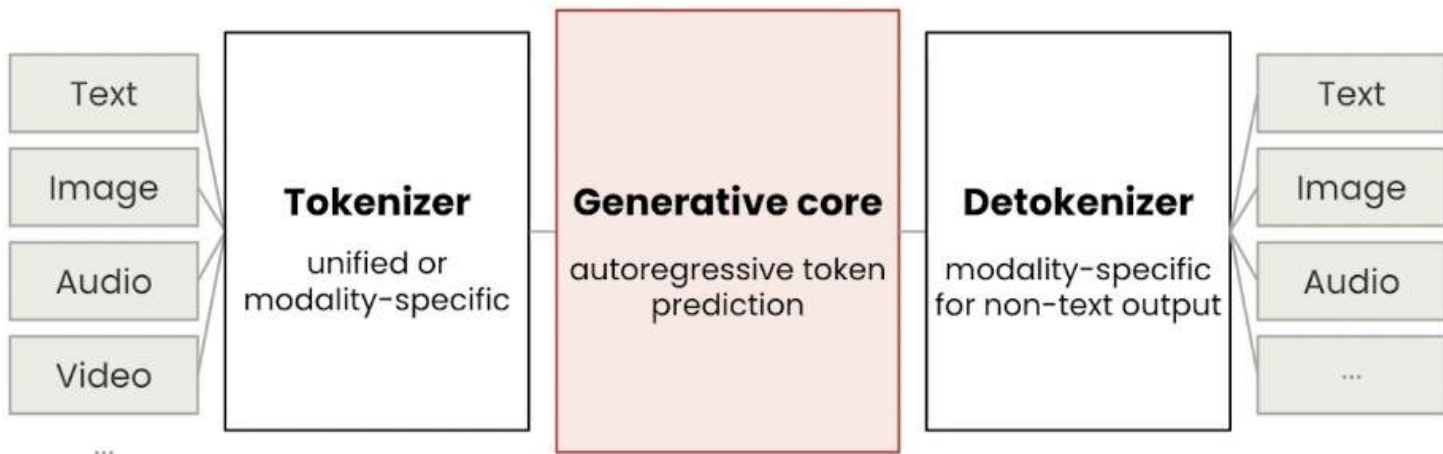
3 Decode

A modality-specific detokenizer converts the token sequence back into the target media.

The generate loop: step 2 zoomed in



T5 joins sequence. Model reads all tokens to predict T6. T = image patch, audio frame, or video token.



Tokenizer architecture varies by model. Active outputs depend on the model and task.

Generative media: diffusion-based generation

Illustrative example of a generic diffusion-based architecture for generative media

1 Condition

Input modalities are encoded into conditioning signals that guide the generation direction.

2 Denoise

Starting from random noise, a denoising network iteratively removes noise guided by conditioning. Repeated dozens to thousands of times.

3 Decode

The final denoised representation is decoded back into the target media.

Prompt: "a sunset over the ocean"

conditions the denoising network at every step



Pure noise

$t = 1.0$



Structure emerging

$t = 0.75$



Shapes clearer

$t = 0.5$



Almost clean

$t = 0.2$



Clean output

$t = 0.0$

Modern models operate in a compressed latent space: denoising happens on latent representations, not raw pixels or audio. A decoder converts the result back to media.

Generative media: hybrid generation

At a high level, many modern models combine both architectures into a single pipeline



Paradigms are combined, the degree of integration varies by model and architecture

Image generation capabilities

Multi-Image composition

Uses multiple images as elements to fuse and create new images

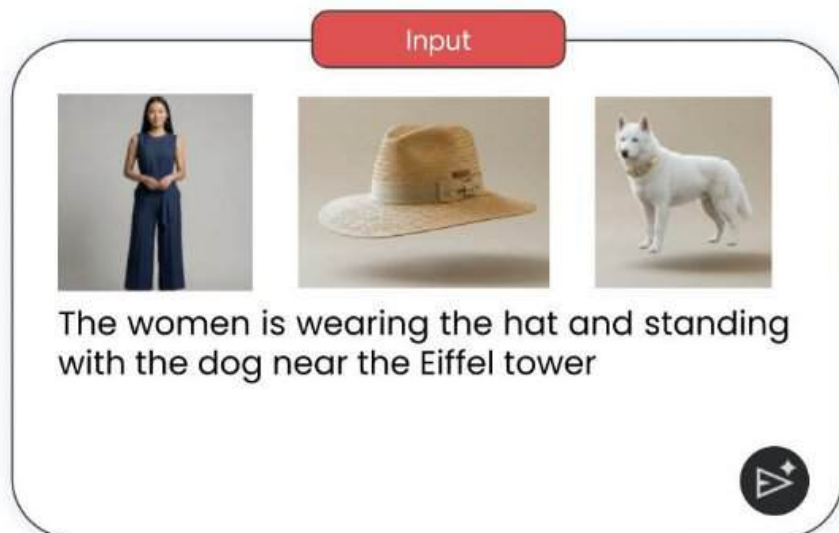


Image generation capabilities

Multi-turn generation

Use natural language to create and refine while maintaining visual consistency across iterations

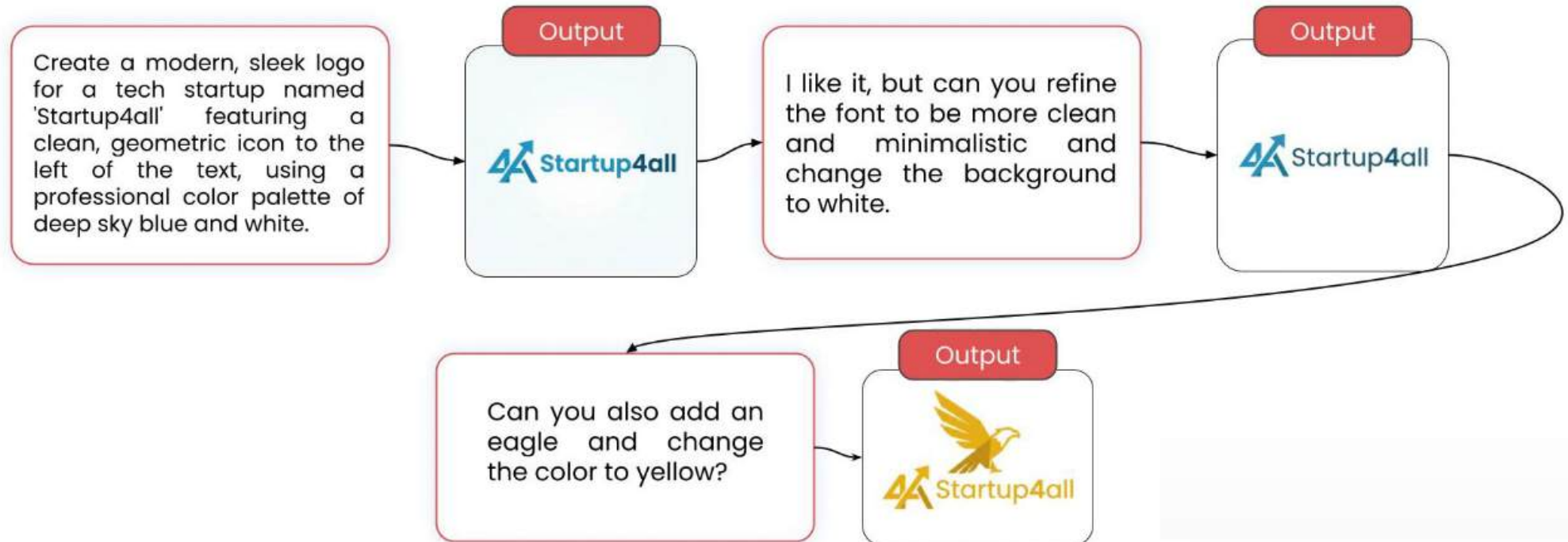


Image generation capabilities

Text rendering

Accurately render long text for titles, descriptions, instructions and more

Input

Generate an image of a poster for a tech conference titled "Future Tech Summit."

The text should also include:

- Date: August 30, 2025
- Register online
- Speakers: Industry leaders in AI, robotics, and innovation



Output

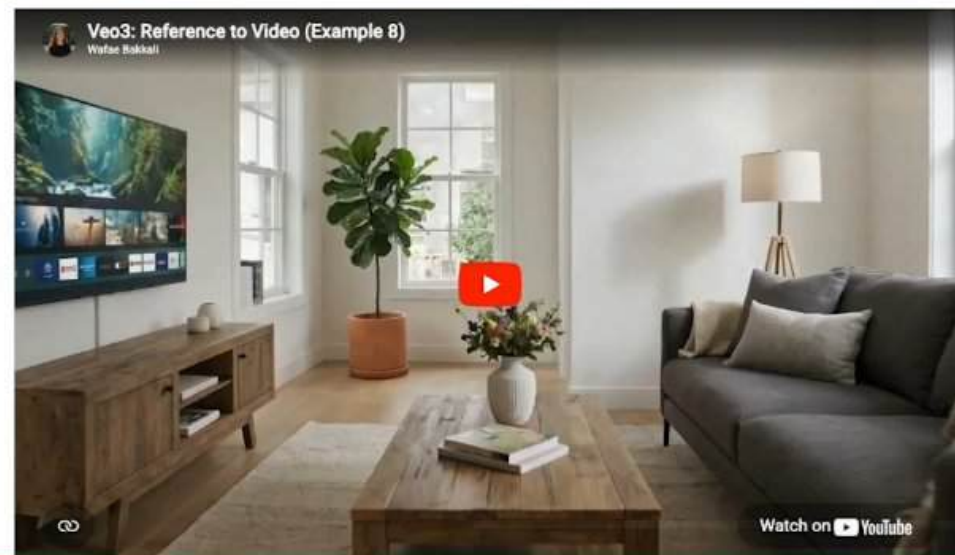


Video Generation Capabilities

Reference to video



Prompt: Home tour based on the **2D floor plan from image 1**. The camera moves from the living room to the bedroom. The living room is furnished with the exact set of furniture shown in the **collage from Image 2**. The bedroom is styled with the bed and furniture from **the collage in Image 3**. All items are placed according to the layout in the plan.

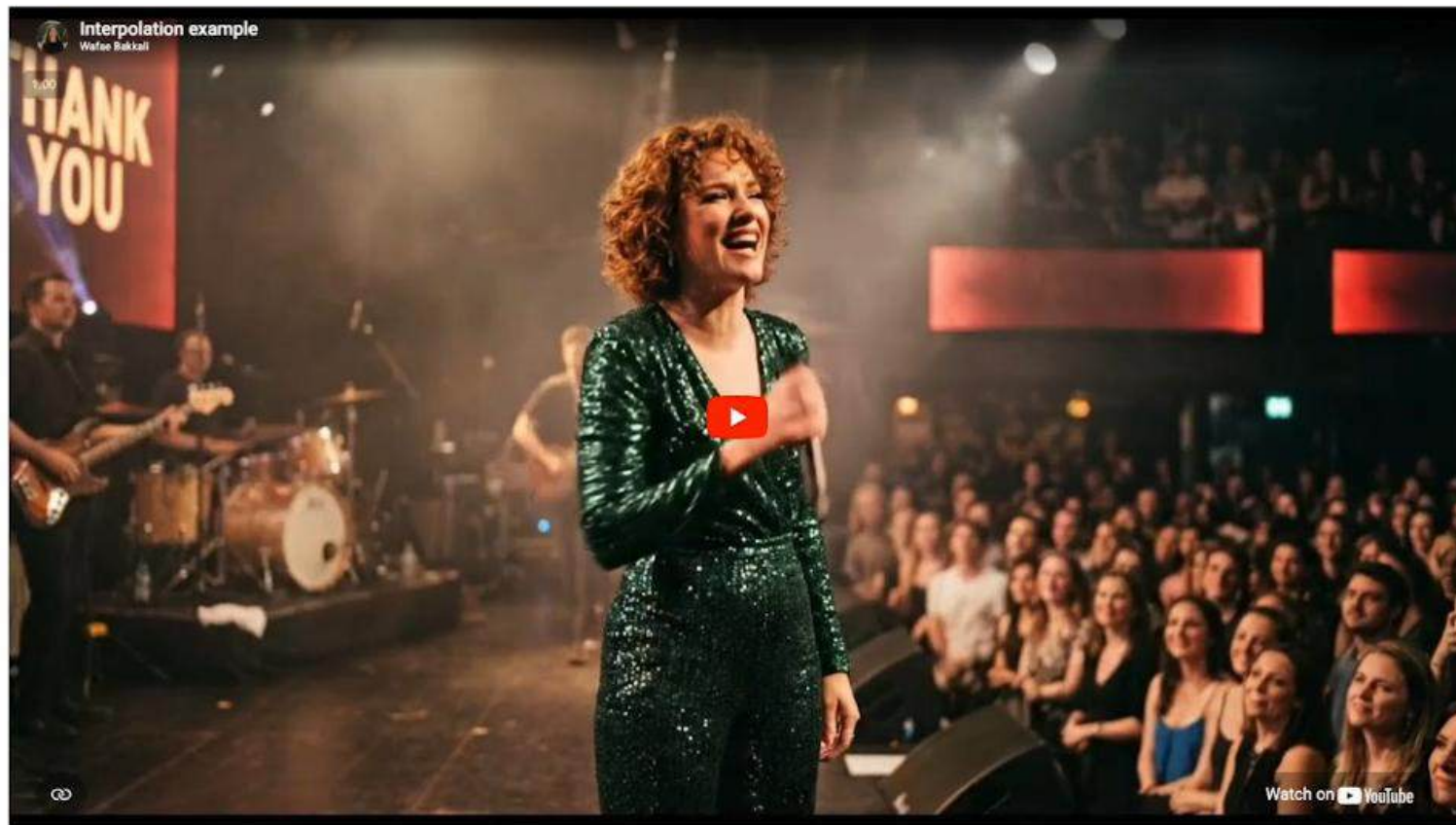


Video Generation Capabilities

First frame & last frame



Prompt: A smooth 180-degree arc shot, starting with the front-facing view of the singer and circling around her to seamlessly end on the POV shot from behind her on stage. The singer sings "Tonight I will keep dreaming, I will keep singing tonight."



Generative Media on Google Cloud



Gemini Image (Nano Banana)

Image generation and editing natively built into the Gemini model.



Veo

Dedicated video generation model with synchronized audio.